

From Prediction to Inference

Quantifying Uncertainty in Model Evaluation

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The question that started it all

In 2020, a clinical collaborator was studying patients with pulmonary embolism. The question was whether four markers of right ventricular dysfunction and cardiac strain added useful predictive information:¹

GDE score

Elevated troponin

Elevated natriuretic peptide

RV:LV ratio > 1.0

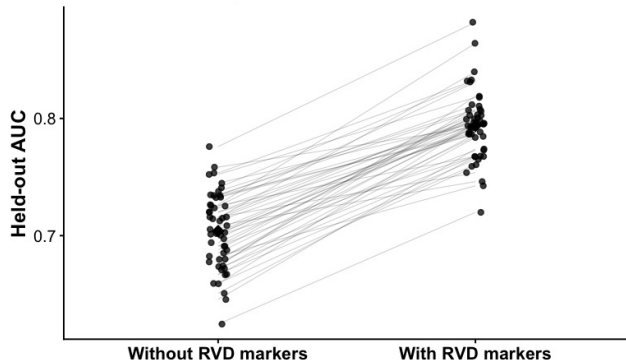
**Do these markers improve prediction of
clinical deterioration?**

And if they appear to improve prediction in our dataset,
how do we know the improvement is real?

¹Weekes, O'Connell et al. (2022). *Critical Care*, 26:160.

The estimated AUC also depended on the random split

50 random train/test splits of the same dataset



What I did:

- 1 Randomly split the same dataset into training and test sets
- 2 Fit both random forests on the training set
- 3 Evaluate both AUCs on the held-out patients
- 4 Repeat 50 times

Average held-out AUC across splits

Without RVD markers: **0.70**

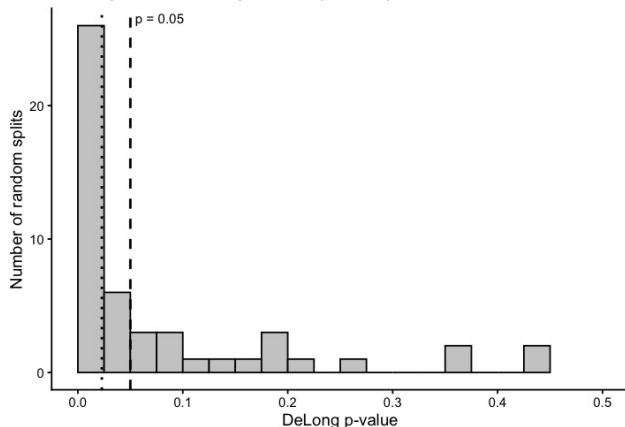
With RVD markers: **0.79**

The estimated performance moved depending on the random split.

Then the inferential answer moved too

Same data, same question, different random splits

Median p-value = 0.023 | 64% of splits had $p < 0.05$



Next step:

For each split, I used **DeLong's test** to compare the two held-out ROC curves.

Across the same 50 random splits:

Median $p = 0.023$

Range: 1.3×10^{-5} to 0.45

So the very same dataset could give

- very strong evidence of improvement, or
- essentially no evidence at all, depending only on the random split.

That ambiguity is what motivated everything that followed....

What uncertainty was I actually supposed to quantify?

Two cultures, still separated 25 years later

Breiman, *Statistical Modeling: The Two Cultures* (2001)

Data modeling culture

Specify a probabilistic model



Estimate parameters



SEs, confidence intervals, p -values

Algorithmic modeling culture

Fit a prediction algorithm



Predict held-out observations



Report predictive performance

**This is how most of us in this room
were trained to do statistics.**

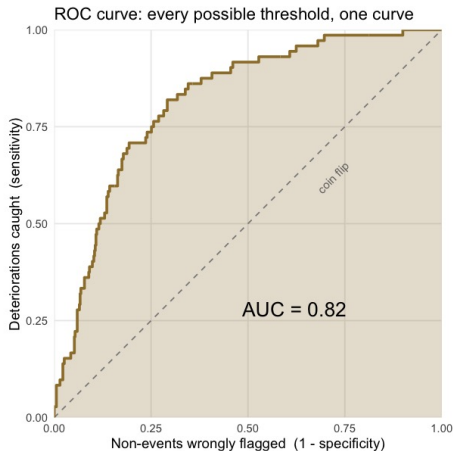
**This is where modern ML and prediction
modeling largely developed.**

Both report numbers calculated from data, but only one culture routinely asks how uncertain those numbers are.

The first question:

**What is a prediction model's
summary measure even measuring?**

AUC: area under the receiver operating characteristic curve



A prediction model gives every patient a score: usually a predicted risk.

AUC, put simply:

Pick 1 patient who had the event and 1 who did not.

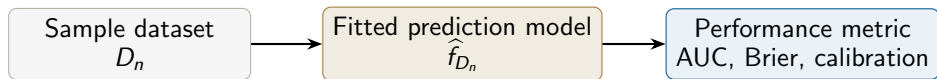
The AUC is the probability the model gives the higher score to the patient who had the event.

- AUC = 0.5: no better than chance
- AUC = 1.0: perfect ranking
- AUC = 0.80: the event patient is ranked higher 80% of the time

AUC is the primary example we use.

but the concepts discussed to any other metric.

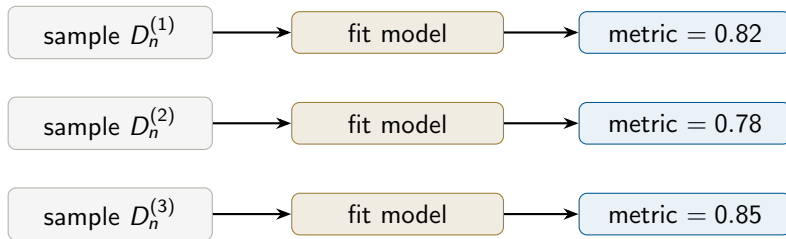
What does a prediction performance metric summarize?



- A prediction model is a learned representation of how outcomes relate to patient characteristics.
- A performance metric summarizes how well that representation works.
- It is not an inherent property of the population alone.
- It is a property of a model's representation of that population.

The metric is a summary of the fitted model's projection of the parent population.

What does the variance of that metric mean?



Same population.
Same sample size.
Same modeling approach.
Different sample.

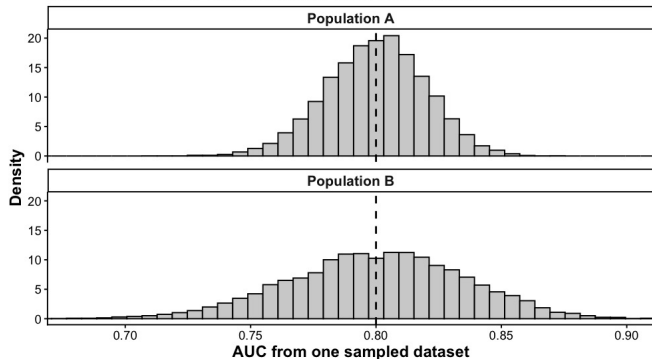
The variance asks how much the reported performance metric would move if we repeatedly drew new samples of size n and fit the same kind of model.

- Each sample gives a different fitted representation.
- Each representation can have different performance in the population.
- That movement is the uncertainty we need to quantify.

The same AUC can carry very different uncertainty

Same average AUC, different uncertainty

Repeated samples of $n = 300$ from two populations with the same event rate and same outcome relationship



Two populations

- Same sample size: **300**
- Same event rate: **20%**
- Same outcome relationship
- Same average AUC: **0.80**

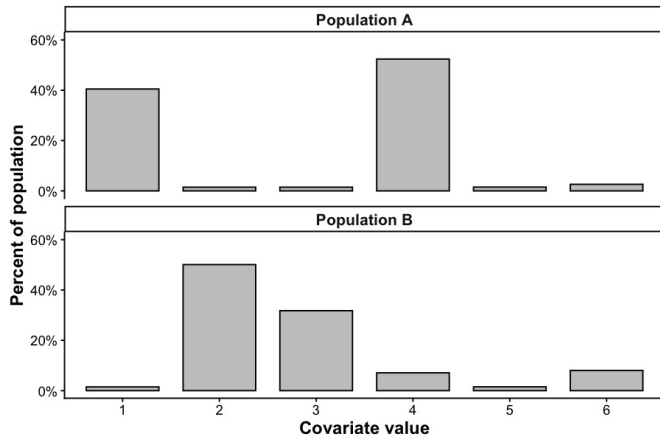
But the AUC from a finite sample is much more stable in one population than the other.

**Same point estimate.
Different level of evidence.**

What changed? Only the covariate distribution

The only thing that changed was the covariate distribution

Both populations have the same event rate, same outcome relationship, and same average AUC.



Held fixed:

- Same sample size
- Same event rate
- Same outcome relationship
- Same average AUC

Changed:

Patient covariate distribution; That alone changed how stable the performance estimate was across repeated samples.

Same average performance. Different uncertainty.

Question:

Why do we care?

Clinical research	Who deteriorates, who is readmitted, who responds to treatment?
Epidemiology	Who is high risk, who should be screened, where are exposures concentrated?
Implementation science	Which sites will adopt, which patients will receive an intervention, where will implementation fail?
Health policy	How does this prediction model differentially impact different populations? How should resources be allocated when decisions depend on predicted risk?

Every one of these models eventually gets summarized by a number:

AUC, MSE, C-index, sensitivity, specificity, Brier score, concordance, etc..

If the summary metric supports a scientific claim, it needs a measure of uncertainty.

Model Evaluation:

What is Cross-Validation, and why does it matter?

To evaluate prediction, we need patients the model did not see

The goal is not performance on the training data.

The goal is generalized performance on new patients from the population of interest.

External validation

Train on one dataset



Test on another dataset



Best direct evidence

Another site, time point, or cohort.

Internal validation

Start with one dataset



Generate held-out predictions



Estimate generalized performance

Used when an independent dataset is not available.

Cross-validation is the standard form of internal validation:

Repeatedly fit the model on part of the data and evaluate it on patients that were held out.

External validation is ideal, but often unavailable, particularly in medical research where data are rare, expensive and/or time consuming to collect.

Clinical machine-learning prediction studies

In a 2023 review of 152 studies, **56%** recommended clinical use with **no external validation**.

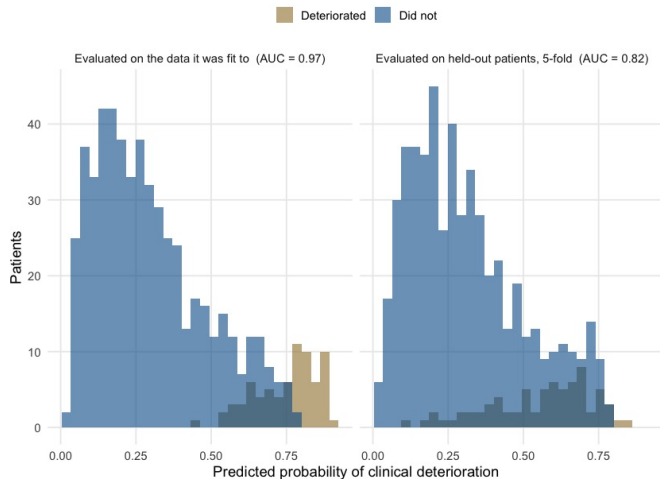
Andaur Navarro et al., *Journal of Clinical Epidemiology*, 2023

TRIPOD calls internal validation **“a necessary part of model development.”**
For many studies, the only available evidence about generalized performance must come from the development dataset itself.

Model Evaluation:

Forms of Cross-Validation

Why we evaluate on patients the model did not see

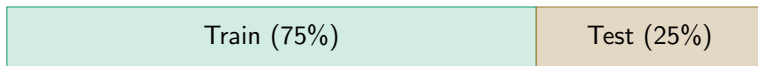


Evaluate on the training data:

- The model has already seen every patient
- Flexible models can partially memorize the sample
- Performance therefore looks better than it really is

Evaluating a model on the data used to train it systematically overstates predictive performance.

The simplest fix: split the data once



Why it works

- Fit the model on one group
- Evaluate it on patients the model never saw
- The test set gives an honest performance estimate

Why it is unsatisfying

- The result depends on one arbitrary random split
- Only part of the data is used to train the evaluated model
- A different split can give a different answer

This is split noise: variability introduced by the cross-validation assignment itself.

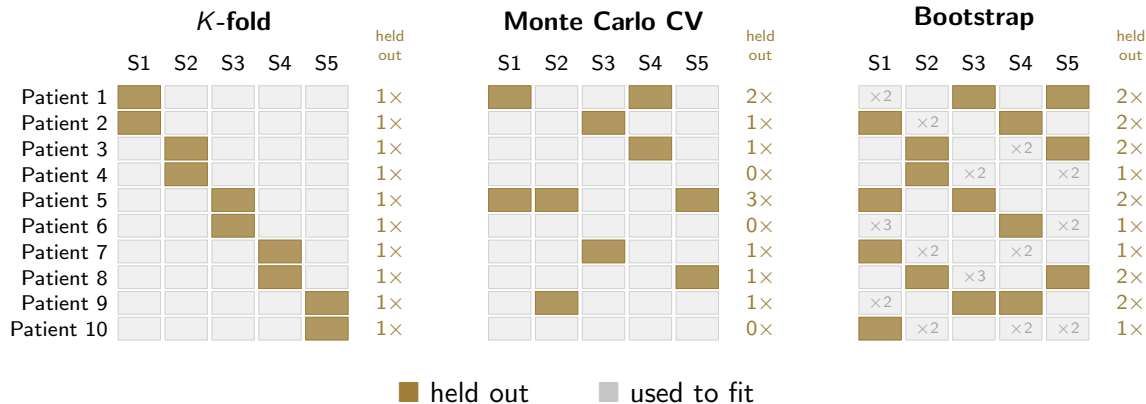
Cross-validation: let every patient take a turn being held out

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Patient 1	held out	fit	fit	fit	fit
Patient 2	held out	fit	fit	fit	fit
Patient 3	fit	held out	fit	fit	fit
Patient 4	fit	held out	fit	fit	fit
Patient 5	fit	fit	held out	fit	fit
Patient 6	fit	fit	held out	fit	fit
Patient 7	fit	fit	fit	held out	fit
Patient 8	fit	fit	fit	held out	fit
Patient 9	fit	fit	fit	fit	held out
Patient 10	fit	fit	fit	fit	held out
Held-out AUC	AUC_1	AUC_2	AUC_3	AUC_4	AUC_5

$$\widehat{AUC}_{CV} = \frac{1}{5} (AUC_1 + AUC_2 + AUC_3 + AUC_4 + AUC_5)$$

Every patient is evaluated by a model that was fit without them, exactly one time.

There is more than one way to cross-validate



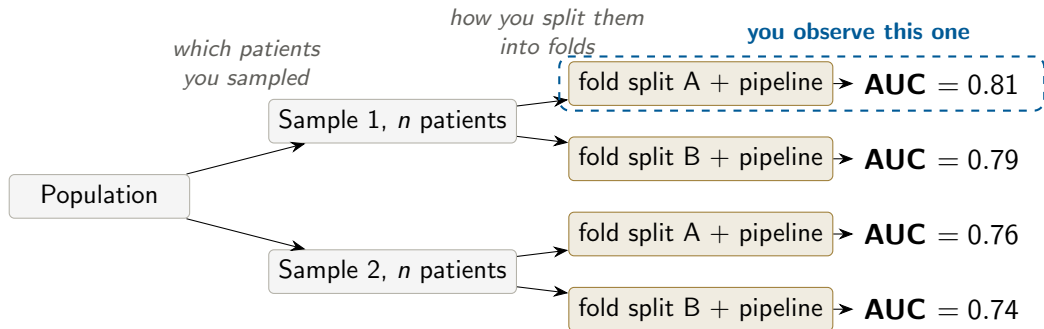
The key difference is how often each patient is used for validation and for training.

What does the cross-validated number actually represent?

Not the performance of the one model you fitted.

It estimates the **average performance of the modeling procedure**: apply the same pipeline to datasets like yours, fit the model, and evaluate it on future patients.

Bates, Hastie & Tibshirani, 2024



The uncertainty has to reflect both.

So which cross-validation design should we use?

- “Leave-one-out ... exhibit[s] a relatively high variance.” Cawley & Talbot, 2007
- “Use 5- or 10-fold” as a practical bias–variance compromise. Hastie et al., 2001; Kim, 2009
- Repeated holdout / Monte Carlo CV can reduce variability by averaging across random splits. Kim, 2009
- Repeated K -fold or MCCV preferred over simple K -fold in small samples. Song et al., 2021
- Larger K can mean **higher variance**. Abedin, Xu & Uddin, 2026
- Fixed choices such as $K = 5$ or 10 may not be universally optimal. Abedin, Xu & Uddin, 2026

Conflicting advice persisted for decades.

Most answers came from model- and dataset-specific empirical comparisons.

I tried to settle it empirically too (JSM 2022).

Estimated from large public datasets ($N > 10,000$); draw samples of $n = 100, 300, 500$; fit a random forest; compare each CV estimate to the truth on the remaining data; 100 repetitions.

$n = 100$	Bias	SD of bias
Single split sample	0.025	0.128
10 $\times$ split sample (Monte Carlo CV)	0.020	0.062
10-fold CV	0.050	0.056
10 $\times$ 10-fold CV	0.048	0.047

- Single split: worst variability everywhere
- Monte Carlo vs. K -fold: the answer flipped from dataset to dataset
- At $n = 500$: no clear winner

Conclusion I wrote at the time: *“an aggregate best method may not exist.”* What I should have written: *I need a theory.*

1,875 studies comparing model performance by cross-validation (Zeng et al., 2026)

- **54%** reported no statistical test and no confidence interval
- Of the assessable tests, **97%** used a method that treated folds as independent

152 clinical machine-learning prediction studies (Andaur Navarro et al., 2023)

- Discrimination had **no precision estimate** in **65%** of main texts and **75%** of abstracts

Most performance estimates are reported either without uncertainty or with an inappropriate one.

19%

false-positive rate
from a nominal 5% test
after one 10-fold CV run

Zeng et al., 2026

77%

false-positive rate
when 10-fold CV was
repeated 30 times

Zeng et al., 2026

Bates et al. (2024): conventional intervals can miss the target at **2–3 times the nominal rate**.

Bengio & Grandvalet (2004): there is no assumption-free unbiased estimator of the variance from the fold estimates alone.

To answer the clinician — *do these variables help, in the population?* — from one dataset, you need:

- ① A **variance** for a cross-validated performance estimate: what it is and what drives it
- ② A way to **estimate** that variance from the single dataset you have
- ③ A **test** for the difference between two procedures

And, before all that, an answer to the design question: **which cross-validation, and what K ?**

None of these existed in general.

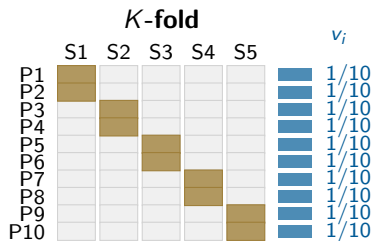
The rest of the talk is about building them from a theory of cross-validation variance.

The Theory:

A theory of cross-validation variance, and an estimator

A cross-validated estimate is a weighted sum of patient contributions

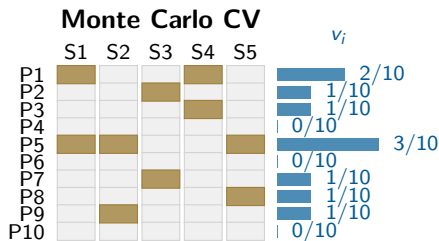
$$\hat{R}_D = \sum_{i=1}^n v_i \ell_i \quad \ell_i = \text{patient } i\text{'s held-out contribution}; \quad v_i = \text{patient } i\text{'s share of all hold-outs}$$



$$\hat{R}_D = \frac{1}{10}\ell_1 + \dots + \frac{1}{10}\ell_{10}$$

$$\sum_i v_i^2 = 0.10$$

equal weights



$$\hat{R}_D = \frac{2}{10}\ell_1 + \frac{1}{10}\ell_2 + \frac{1}{10}\ell_3 + 0\ell_4 + \frac{3}{10}\ell_5 + \dots$$

$$\sum_i v_i^2 = 0.18$$

unequal weights

Imbalance concentrates influence. This makes the squared weights larger.

Every patient influences the estimate twice

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
P1					
P2					
P3					
P4					
P5	fit	fit	held out	fit	fit
P6					
P7					
P8					
P9					
P10					

Validation influence: patient 5 is held out once. If their outcome is surprising and predicted badly, that bad prediction goes directly into the performance estimate. Their validation weight is v_5 .

Training influence: patient 5 is used to fit the four models that predict the other patients. If their outcome is surprising, they can pull those fitted models toward them, changing other patients' predictions. Their training weight is w_5 .

The same surprising patient can affect the estimate through validation and through training.

The variance identity for a cross-validated metric

$$\text{Var}(\hat{R}_D) = \underbrace{\sigma_V^2 \sum_i v_i^2}_{\text{held-out influence}} + \underbrace{C_0 \sum_i w_i^2}_{\text{model-fitting influence}} + \underbrace{2\gamma_0 \sum_i v_i w_i}_{\text{move together}} + \underbrace{\rho}_{\text{interactions}}$$

- σ_V^2 : how much a patient's **own held-out prediction can move the performance estimate**
- C_0 : how much a patient can **change the fitted models**, and therefore other patients' predictions
- γ_0 : whether those two influences tend to **reinforce or offset one another**
- $\rho \geq 0$: higher-order interactions among patients

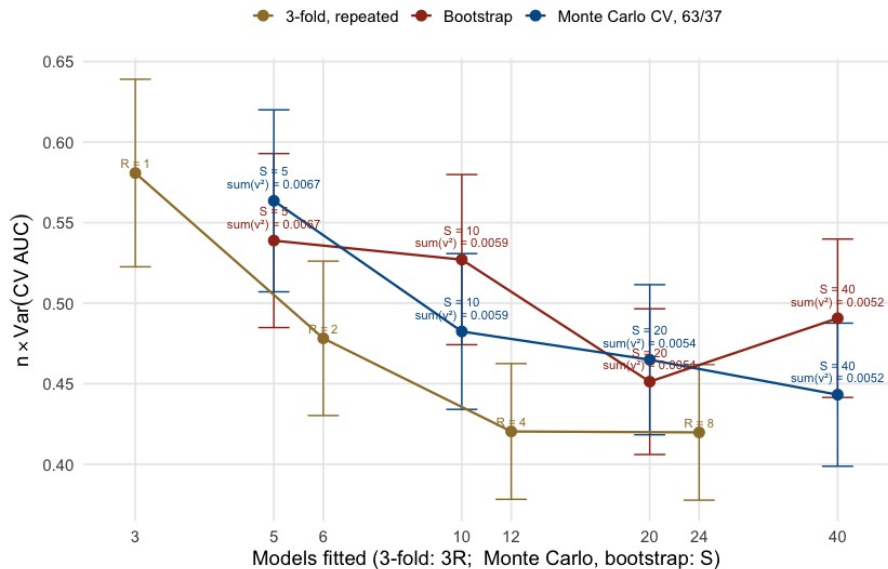
The influence terms depend on the population, model, and metric. The weights v_i and w_i are the only place the cross-validation procedure influences the variance itself.

Balanced K -fold minimizes variance

- In K -fold, **every patient gets the same say** in validation and in training.
- In an unbalanced design, some patients get more influence than others.
- Concentrating influence on fewer patients makes the performance estimate **less stable**.

For any model, any sample size, and any standard CV metric.

The variance tracks the weights, exactly as the identity says



Repeating the design reduces split noise

- Even with the **same patients**, redrawing the splits can change the reported performance.
- K -fold has **less split noise** because balance fixes the patient weights across partitions.
- Monte Carlo CV has more split noise because the weights themselves change from one set of splits to another.
- Repeating either design and averaging reduces the remaining split noise.

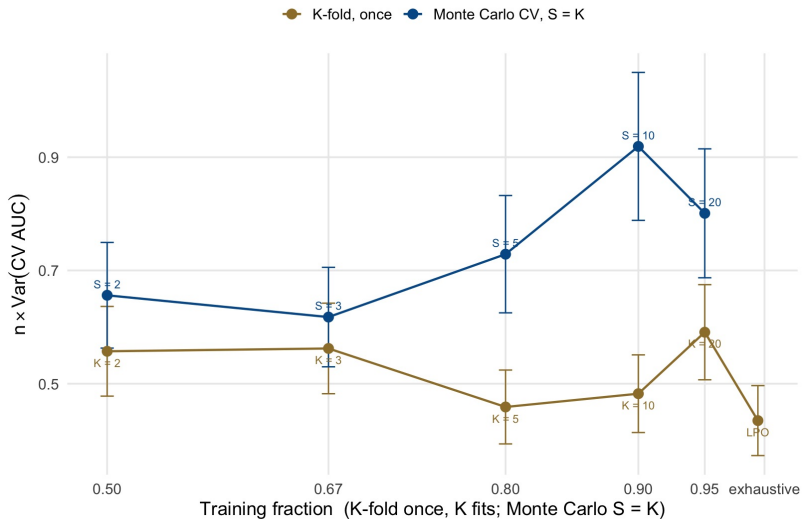
Repetition reduces split noise — but not sampling uncertainty.

More folds do not mean higher variance

- More folds do create **more overlap** between the training sets.
- But K -fold remains **perfectly balanced** at every K : each patient gets the same overall validation and training weight.
- So increasing K does **not increase the leading variance terms**.

K changes training size, bias, and split noise — not the leading variance.

Same fits, same training fraction: balance versus the draw



$$\text{Var}(\hat{R}_D) = \sigma_V^2 \sum_i v_i^2 + c_0 \sum_i w_i^2 + 2\gamma_0 \sum_i v_i w_i + \rho$$

- Most conventional approaches quantify uncertainty in the **held-out predictions**.
- But the patients also influenced the **models that generated those predictions**.
- That model-fitting influence, and its relationship with held-out influence, is missing.

The held-out predictions are only part of the uncertainty.

Estimation

**How do we estimate the variance
from one dataset?**

No universally unbiased estimator of K -fold CV variance exists.

Bengio & Grandvalet, 2004

So any estimator has to impose structure somewhere.

The question is: which assumption makes the most sense?

- **Modified bootstrap:** resample patients with replacement, rerun cross-validation, and rescale with a fixed correction factor. The learner is fit to datasets with duplicated patients, and the factor is calibrated to one training fraction. Cai et al., 2025
- **Nested cross-validation:** a fresh outer holdout, then extrapolate from a smaller training size. Targets the performance of *this* fitted model rather than the procedure, at roughly ten times the computation. Bates, Hastie & Tibshirani, 2024

Both impose structure to get around the Bengio–Grandvalet impossibility result. The question is whether the structure is placed where the identity says the variance lives.

The identity suggests where to put the assumption

The variance identity tells us what has to be recreated:

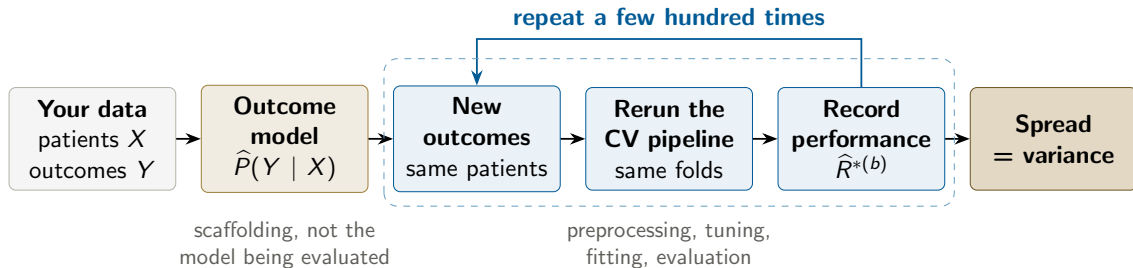
- A patient's outcome affects their **own held-out performance**.
- The same outcome affects the **models trained on that patient**.
- Those two effects can **move together**.

So instead of modeling a missing variance component or extrapolating from a smaller training size:

Model how outcomes depend on the patient characteristics we observed, then generate new outcomes and rerun the entire prediction procedure.

The assumption becomes: can we generate realistic new outcomes for these same patients?

Keep the patients. Keep the folds. Regenerate the outcomes. Rerun everything.



One small extra term covers *which patients* were sampled; it cancels when two models are compared.

Simulating from a fitted model to get error bars is what statisticians have always done.

PASR does it for the whole prediction pipeline.

One assumption, four guarantees

Everything rests on the outcome model standing in for the truth. What we can prove:

Unbiased under the outcome model

At any number of reruns.
The *only* systematic error is the outcome model's, and it is bounded.

Chooses the outcome model correctly

Cross-validated Brier score picks the candidate closest to the truth in exactly the distance that matters.

Errs wide, never narrow

If the outcome model cannot match how predictable the outcomes are, the interval gets *conservative*.

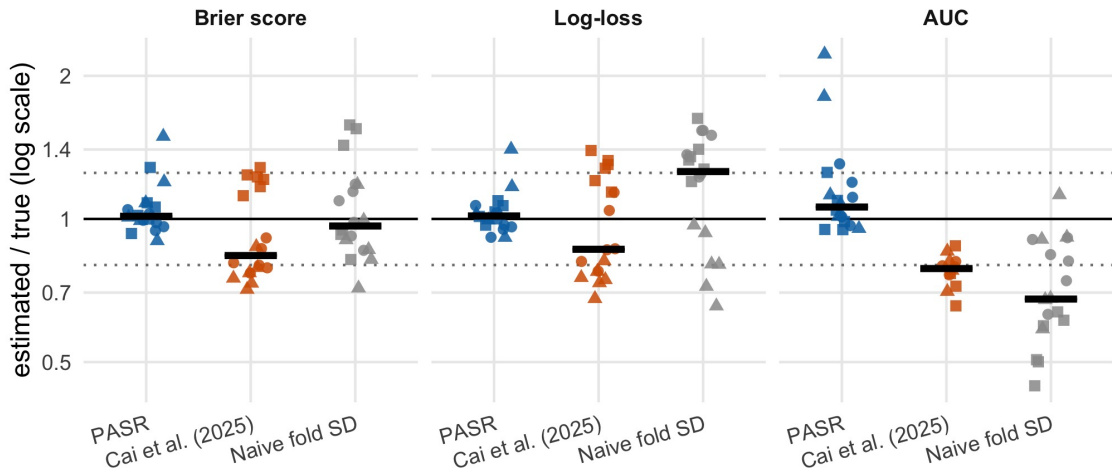
Checks itself

Performance on synthetic vs. real outcomes:
a gap flags a conservative interval.

A visible assumption, a proved error bound, a built-in diagnostic.

Estimated variance relative to the true variance of the CV estimate

Each point is one setting (truth x learner x n); bar = median; 1 = unbiased



Does it recover the truth?

18 settings: two simulated populations $\times$ logistic, random forest, k -NN $\times n = 250, 500, 1000$; three metrics (Brier, log-loss, AUC). Truth from 1,000–2,000 independent datasets per setting.

	estimated variance / true variance					within 10%
	10th	25th	median	75th	90th	of truth
PASR	0.95	0.97	1.02	1.09	1.27	42/54
Cai bootstrap	0.73	0.77	0.81	1.12	1.25	2/48
Naive fold SD	0.62	0.76	0.92	1.22	1.48	13/54

Percentiles over the 54 setting $\times$ metric combinations; 1 = unbiased.

- PASR's misses are one family (interaction population, random forest, $n \leq 500$): it errs *high*, up to $2.2\times$, and the diagnostic flagged it
- The bootstrap errs *low* in 34 of 48 settings; its AUC estimate was undefined in most datasets at $n = 250$
- The naive error bar is off by more than 25% in half the settings, in both directions
- About 3,000 model fits per PASR estimate vs. 8,000 for the bootstrap

Implications

From Prediction to Inference

Cross-validation is now an inferential procedure

Where we started

Fit a prediction algorithm



Cross-validate



Report a number

A summary statistic with no sampling distribution attached to it.

Where we are

Fit a prediction algorithm



Cross-validate



Variance, interval, test

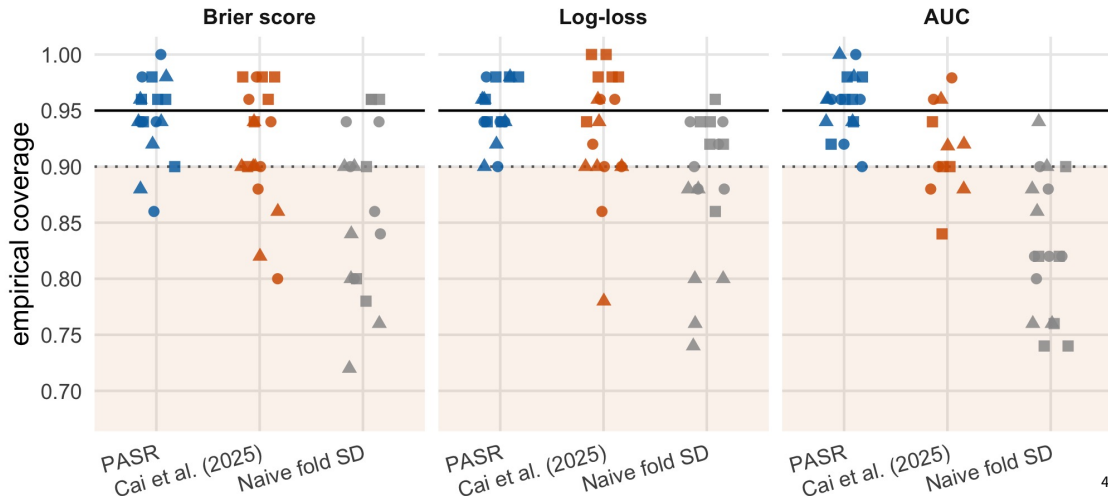
A statistic with a known variance, an estimator for it, and a test.

- The reported performance becomes a **point estimate with a confidence interval**
- “Model A beats model B” becomes a **hypothesis test**, not a comparison of two numbers
- The design question has an answer: **balanced K -fold, repeated**

Prediction → **Inference**

Coverage of nominal 95% confidence intervals for CV performance

Each point is one setting; shaded region = coverage below 90%



Back to the question that started it all

Do the four markers improve prediction of clinical deterioration — in the population, not just in this sample?

$n = 610$ patients, 72 events. Same random forest, same pipeline. Repeated 5-fold CV,

	CV-AUC	95% CI
With RVD markers	0.81	[0.73, 0.88]
Without RVD markers	0.70	[0.61, 0.79]
Difference	0.10	[0.04, 0.17] $p = 0.001$

- ✓ **A variance**, from the identity: what it is and what drives it.
- ✓ **An estimate**, from PASR, using only the dataset we had.
- ✓ **A test** whose p -value means what the clinician thinks it means.

The markers appear to improve prediction with statistical significance.

Looking Forward:

Where this is going

PI O'Connell, co-PI Speiser. The two papers are the preliminary work; the proposal turns them into general inference.

Aim 1. Inference within a population. Four questions, each answered from the variance identity.

1A Inference. *"Is model A better than model B?"*

Intervals and tests for comparing models, valid even when the two are nearly identical, where the normal approximation breaks.

1B Variance partitioning. *"Where does the error bar come from?"*

Split it into its three parts and attribute it to patients, outcome classes, and covariate strata: is it the data, the model, or both.

1C Imbalance. *"How many events do I need?"*

Rare outcomes and stratified folds: how imbalance inflates the variance, and how many events is enough for which metric.

1D Repeats. *"How many times should I repeat CV?"*

How many repetitions of K -fold buy real precision, and when to stop paying for compute.

Aim 2. Inference across populations. *"It works here. Will it work there?"*

- **The setting.** A model trained at one site, evaluated at another site or in a subgroup. Cross-validation as it stands only speaks to the sample it was run on.
- **The estimate.** Reweight the validation so the performance estimate refers to the *target* population, not the source.
- **The error bar.** Add the uncertainty from *estimating* the weights and from the target sample itself, on top of the three terms from Aim 1.
- **The safeguard.** Diagnostics that say when the source data cannot support any claim about the target.
- **The testbed.** Advocate Health: Winston-Salem, Charlotte, Georgia, Midwest .

- 1 **A cross-validated metric is a statistic.** It has a sampling distribution. Report it with an interval, or draw no conclusions from it.
- 2 **Its variance has three parts:** noisy held-out contributions, moving fitted models, and how the two move together.
- 3 **Use K -fold, repeated.** Balanced, so variance-optimal at every n . K does not change the leading variance.
- 4 **The usual error bars see one part. PASR sees all three** by rerunning the whole procedure on synthetic outcomes.

Cross-validation is now an **inferential** procedure, not just an estimation one.



A recipe for better prediction

And for the one lab member who does not like soup



Zero variance. Zero bias. Zero soup.

Thank you

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O'Connell & Speiser. *A Theory of Cross-Validation Variance*. arXiv, 2026.

O'Connell & Speiser. *Estimation of Cross-Validation Variance via Procedure-Aligned Synthetic Resampling*. arXiv, 2026.

O'Connell. *Random Forests as Statistical Procedures: Design, Variance, and Dependence*. arXiv, 2026.